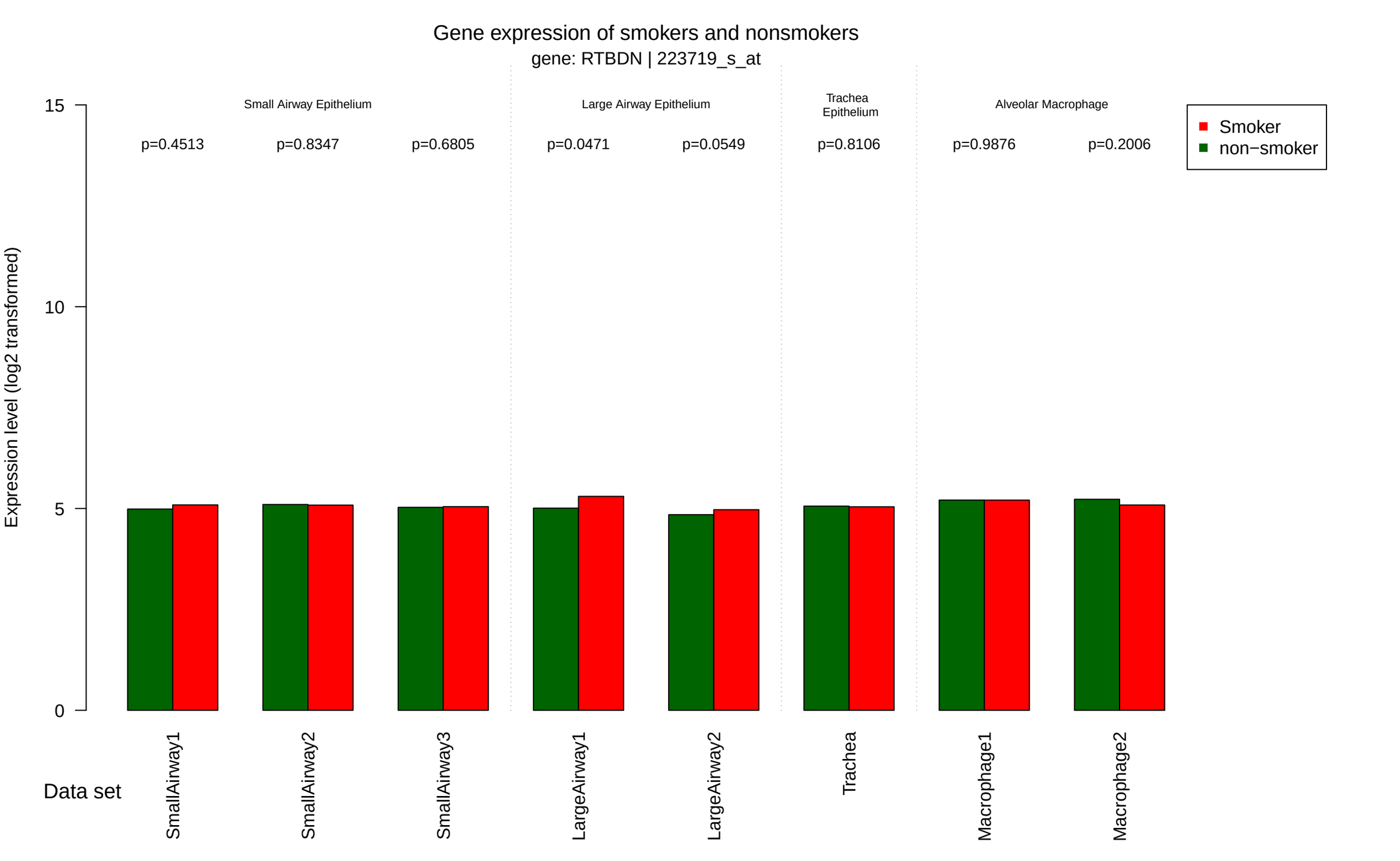
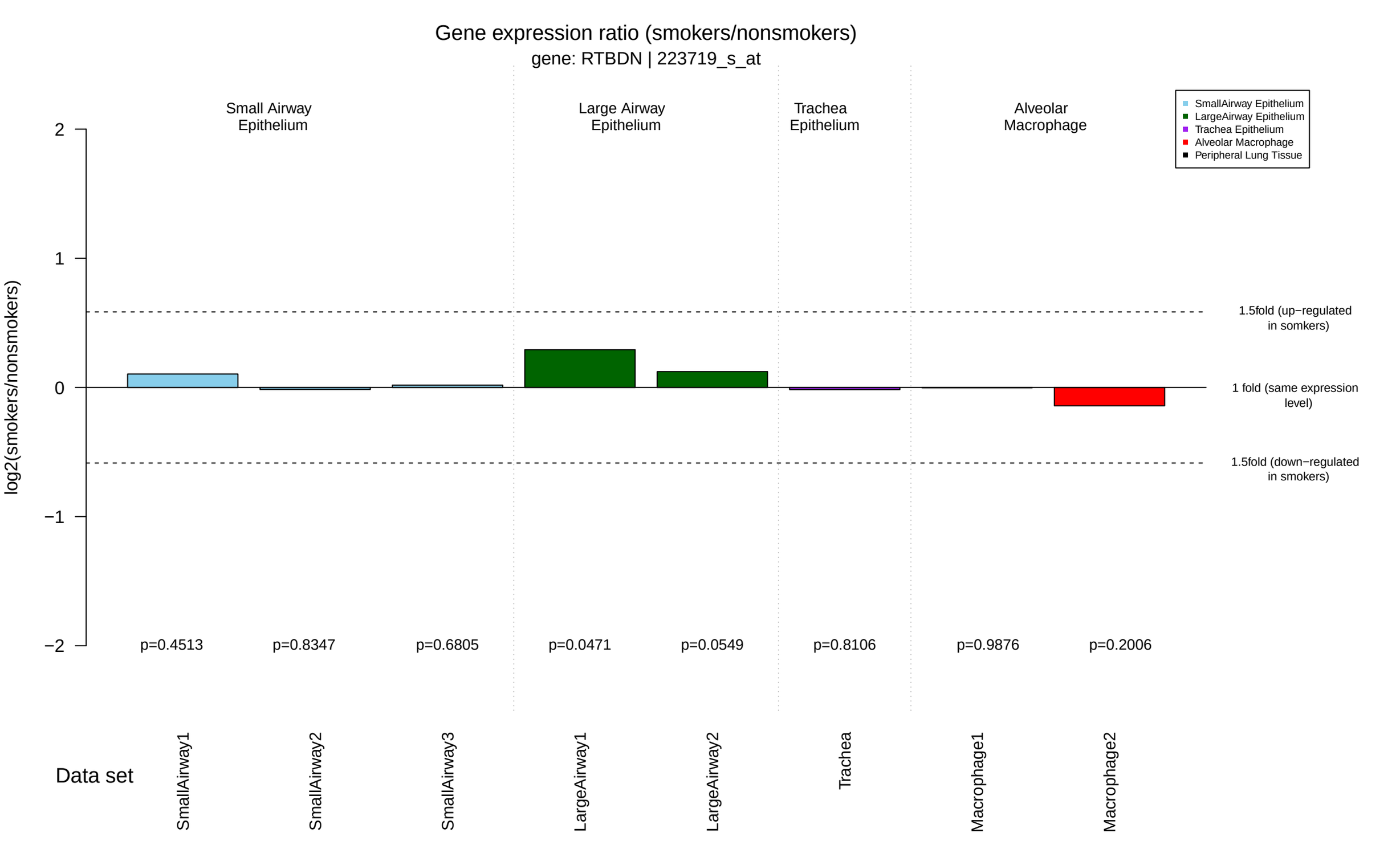
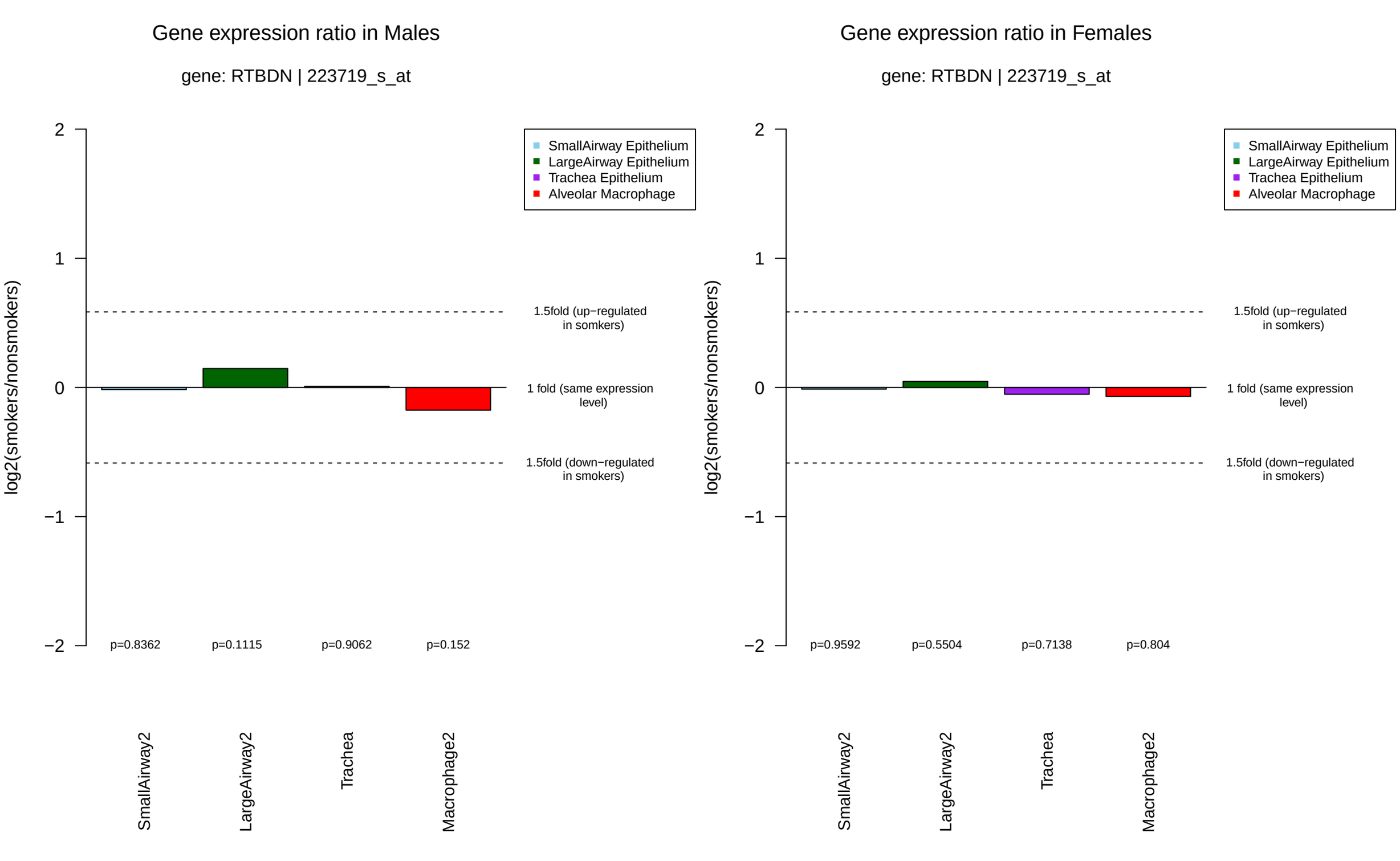
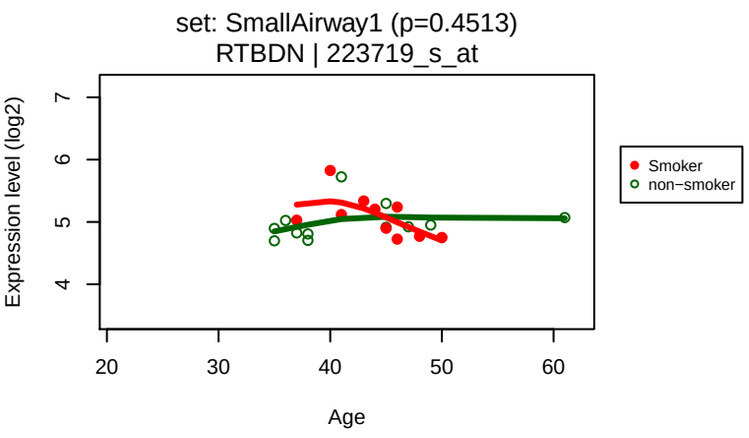
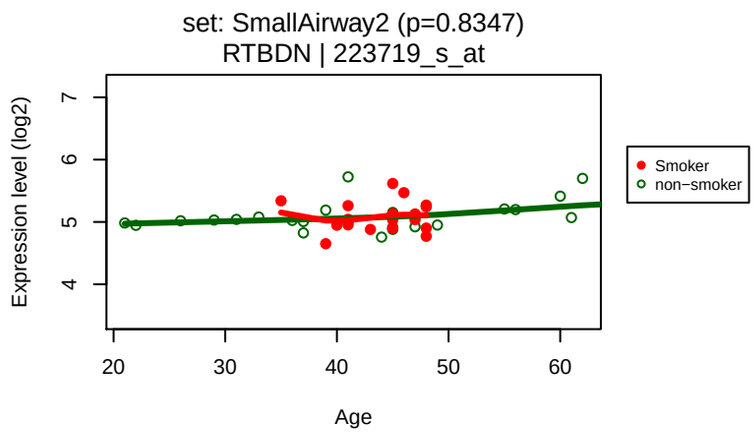
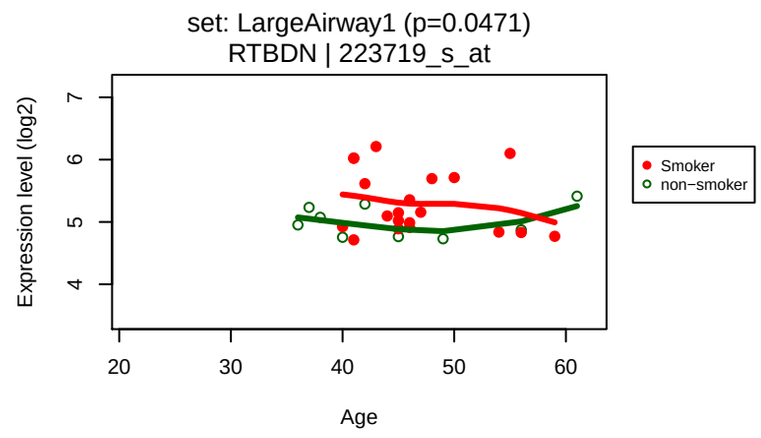

Gene expression of smokers and nonsmokers
gene: RTBDN | 223719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4513
p=0.8347
p=0.6805
p=0.0471
p=0.0549
p=0.8106
p=0.9876
p=0.2006
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

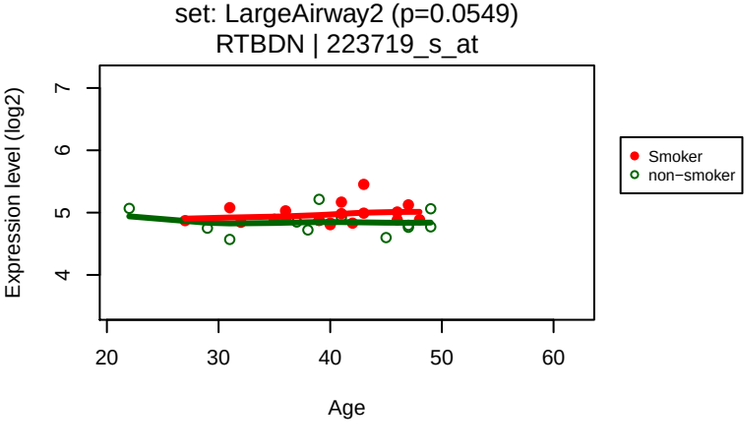

Gene expression ratio (smokers/nonsmokers)
gene: RTBDN | 223719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4513
p=0.8347
p=0.6805
p=0.0471
p=0.0549
p=0.8106
p=0.9876
p=0.2006
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

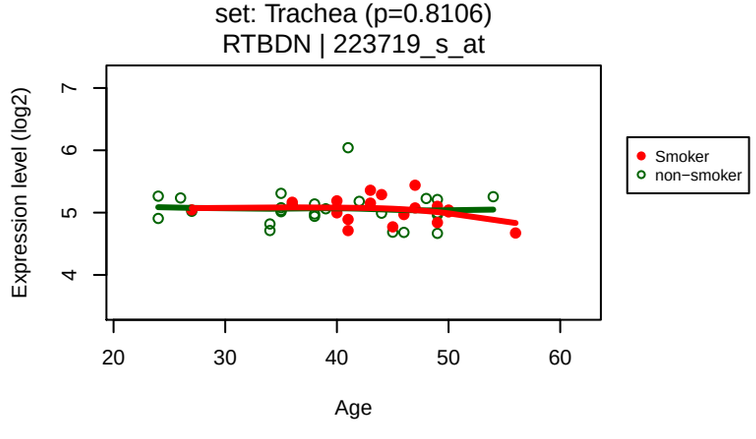

Gene expression ratio in Males
gene: RTBDN | 223719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8362
p=0.1115
p=0.9062
p=0.152
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTBDN | 223719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9592
p=0.5504
p=0.7138
p=0.804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

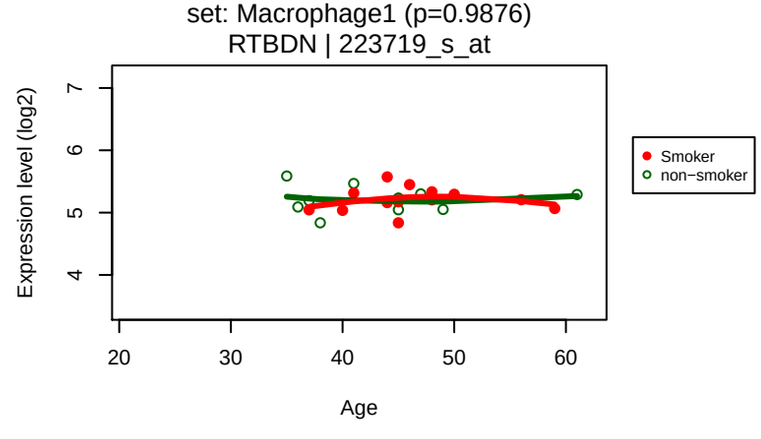

set: SmallAirway1 (p=0.4513)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker

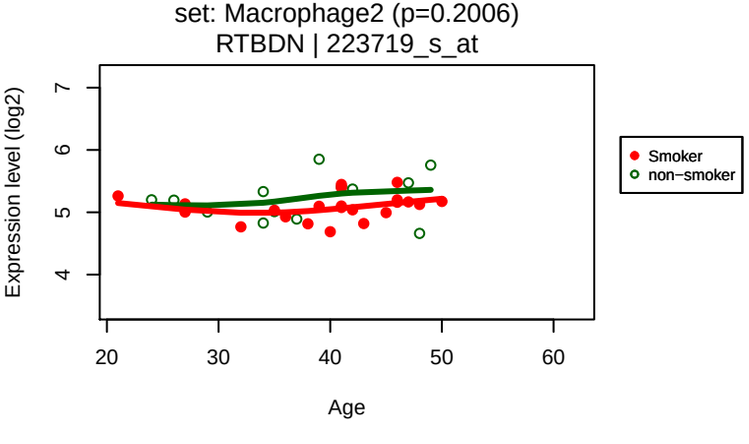

set: SmallAirway2 (p=0.8347)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker

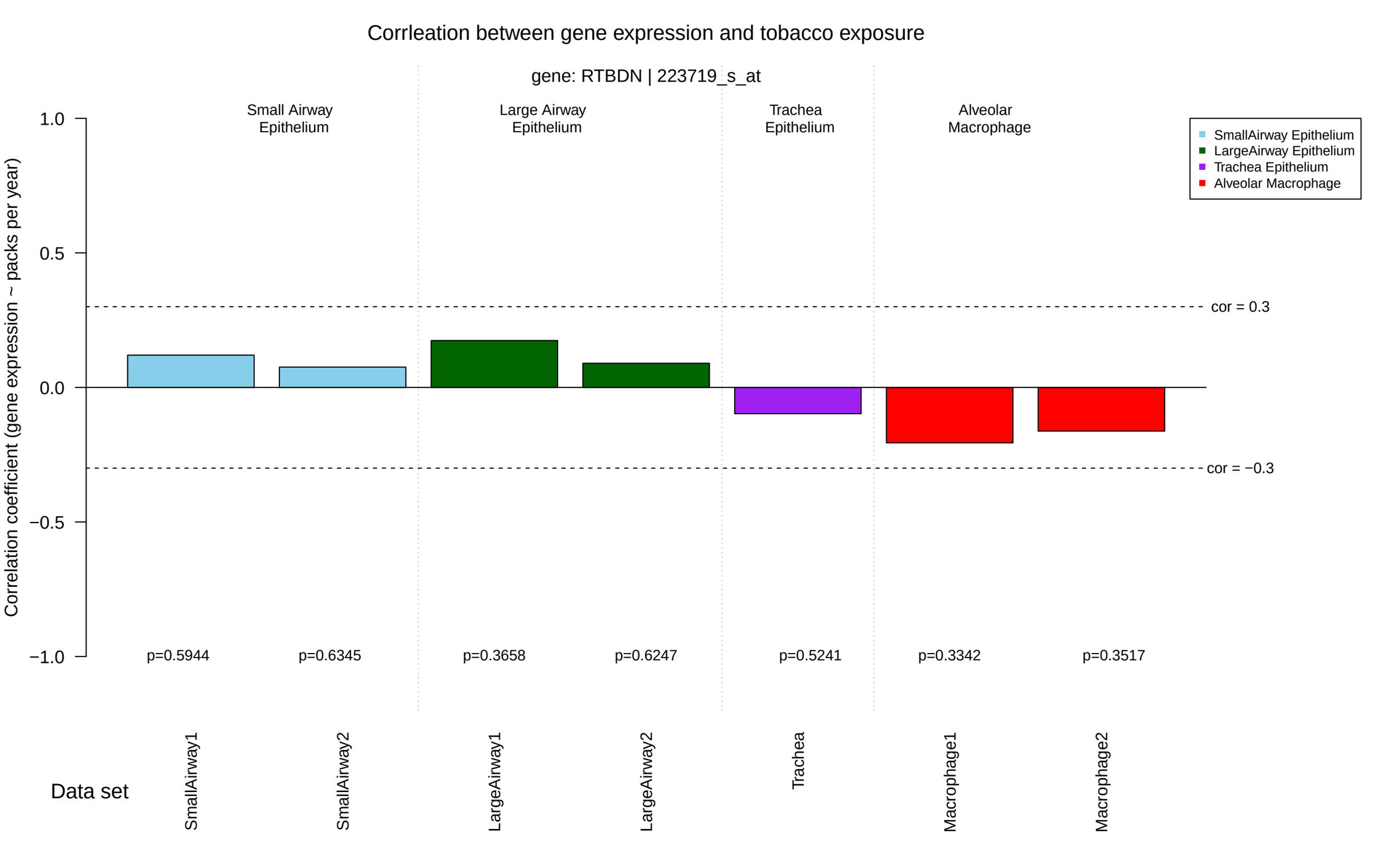

set: LargeAirway1 (p=0.0471)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0549)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8106)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9876)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2006)
RTBDN | 223719_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTBDN | 223719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5944
p=0.6345
p=0.3658
p=0.6247
p=0.5241
p=0.3342
p=0.3517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2